

Database Project Report Template

DueDate:[InsertDate - Week 3-4.]

Project ID & Title: University Research & Lab Asset Tracker

A. Project Identity

- **Team Name:** G1
- **Team Members:**
 - Phạm An (n22dcat002@student.ptithcm.edu.vn)
 - Lâm Minh Chiến (n24dece057@student.ptithcm.edu.vn)
 - Đỗ Vương Bảo Anh (n24dece054@student.ptithcm.edu.vn)
- **Project Title :** University Research & Lab Asset Tracker

B. Report structure

1. Introduction & Project Scope

1.1 System Objective

1.1.1 Problem Statement

Universities operate multiple research labs serving faculty and students, using high-value equipment and chemicals while also receiving funding from multiple research grants provided by different funding organizations. Currently, the management of lab assets, chemical inventory, and lab access rights is often carried out manually using paper records and separate spreadsheets, leading to the following problems:

- It is difficult to control who is accessing hazardous chemicals or restricted areas, creating occupational safety risks.
- Equipment and chemicals are not associated with a specific lab or grant, making them difficult to trace during audits.
- There is no mandatory constraint ensuring that only personnel with appropriate safety certifications are allowed to access hazardous materials.
- There is insufficient data linkage among personnel (faculty/students), research grants, and the laboratory assets they use.

The objective of the project is to design and build a relational database named "University Research & Lab Asset Tracker" for the centralized management of: (1) research grants, (2) laboratory equipment, (3) chemical inventory, and (4) access logs for faculty and students entering or leaving laboratories. The most important integrity constraint is that only personnel with valid, unexpired certifications may access materials or areas with the corresponding hazard level. This constraint will be enforced at the database layer through an SQL Trigger.

1.1.2 Scope

- In scope: management of departments, personnel (faculty/students), safety certifications, laboratories, equipment, chemicals, research grants, and access logs.

- Out of scope: the university's full financial accounting system, academic administration (course registration), and the complete HR system for university employees.

1.1.3 Functional Requirements

- FR1: The system allows an Administrator (Admin) to register new laboratory equipment and assign the equipment to a specific laboratory.
- FR2: The system allows Faculty members to create and manage research grant records, including the funding organization name, grant amount, and project duration.
- FR3: The system records every personnel access event at a laboratory, including the timestamp and access type (entry/exit).
- FR4: The system must validate a personnel member's safety certification before allowing access to chemicals or laboratories classified as hazardous. This rule is enforced by an SQL Trigger, not only at the application layer.
- FR5: The system tracks chemical inventory, including stock quantity, storage location, and hazard classification (hazard class).
- FR6: The system allows multiple personnel members (faculty/students) to be assigned to a grant with specific roles (PI, Co-PI, Research Assistant).
- FR7: The system supports report queries, including lists of unauthorized access attempts, equipment by laboratory, and chemicals with related certifications that are nearing expiration.

1.1.4 Non-Functional Requirements

- NFR1 (Security): Certification data and personnel identification information must be subject to strict access control; encryption at rest is recommended for sensitive fields.
- NFR2 (Performance): Access-log lookup queries used for auditing must return results in under 200 ms with up to 100,000 records.
- NFR3 (Availability): The system should be available 24/7 because laboratories may be accessed outside normal working hours.
- NFR4 (Data Integrity): The rule that "only personnel with valid certifications may access hazardous materials" must be enforced at the database layer (trigger/constraint) and must not depend entirely on the application layer.

1.2. Business Rules

The following business rules form the basis for the ER/EER Diagram and will be mapped directly to foreign-key constraints, primary-key constraints, and triggers in Phases 2 and 3.

1.2.1 Personnel & Certification

- BR1 - Objects: PERSONNEL, FACULTY, STUDENT: Each Personnel has a unique person_id. A Personnel may be specialized as either Faculty or Student. The specialization is disjoint and partial; therefore, a Personnel member may belong to neither subtype.
- BR2 - Objects: FACULTY, DEPARTMENT: Each Faculty member belongs to exactly one Department; a Department may have many Faculty members.
- BR3 - Objects: STUDENT, FACULTY: Each Student may have at most one Faculty member as an advisor; a Faculty member may advise many Students.
- BR4 - Objects: PERSONNEL, CERTIFICATION, PERSON_CERTIFICATION: A Personnel member may be issued zero or more Certifications; a Certification may be issued to many Personnel members (an N:N relationship through PERSON_CERTIFICATION), and each issuance has its own issue_date and expiry_date.
- BR5 - Objects: CERTIFICATION, PERSON_CERTIFICATION: A Certification is valid only when the current date falls within [issue_date, expiry_date]; expired certifications are not considered valid when determining access permission.

1.2.2 Laboratories, Equipment & Chemicals

- BR6 - Objects: EQUIPMENT, LAB: Each Equipment item belongs to exactly one Lab; a Lab may contain zero or more Equipment items.
- BR7 - Objects: CHEMICAL, LAB: Each Chemical is stored in exactly one Lab; a Lab may store zero or more Chemicals.
- BR8 - Objects: LAB, CHEMICAL, PERSONNEL, CERTIFICATION: Each Lab and each Chemical has a hazard_level/hazard_class. If hazard_level != 'None', only Personnel with a valid Certification at the corresponding hazard level or a higher level may access it (this constraint is enforced by an SQL Trigger BEFORE INSERT on ACCESS_LOG).

1.2.3 Research Grants

- BR9 - Objects: RESEARCH_GRANT, FACULTY: Each Research Grant has exactly one Principal Investigator (PI), and the PI must be a Faculty member (not a Student).
- BR10 - Objects: FACULTY, RESEARCH_GRANT: A Faculty member may be the PI of multiple Grants.
- BR11 - Objects: RESEARCH_GRANT, PERSONNEL, GRANT_MEMBER: A Grant may have many participating Personnel members with different roles (PI, Co-PI, Research Assistant) through the GRANT_MEMBER associative table; a Personnel member may participate in multiple Grants.
- BR12 - Object: RESEARCH_GRANT: The Grant amount must be > 0, and end_date must be later than start_date.

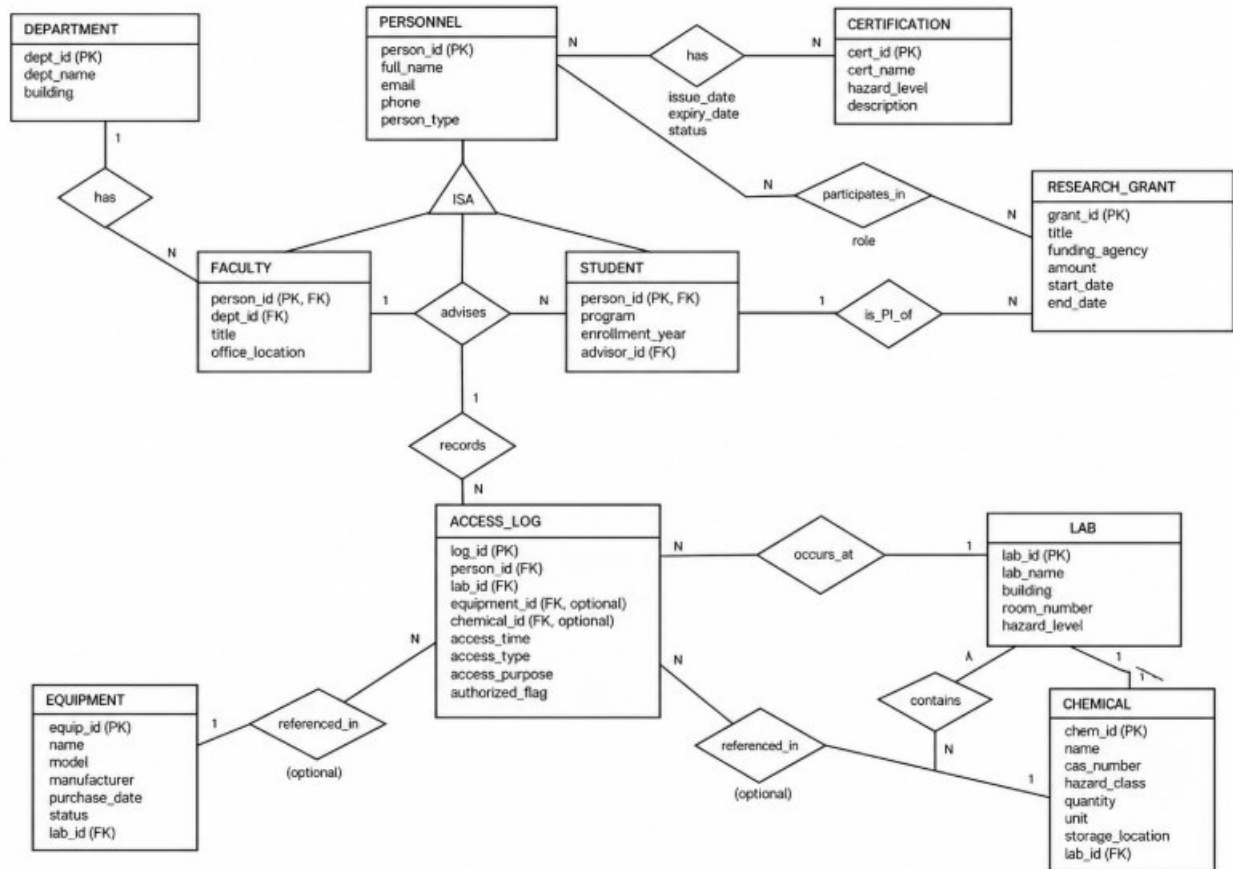
1.2.4 Access Log

- BR13 - Objects: ACCESS_LOG, PERSONNEL, LAB, EQUIPMENT, CHEMICAL: Each Access Log record must reference an existing valid Personnel member and Lab; it may optionally reference a specific Equipment item and/or Chemical if the event relates to that item or substance.
- BR14 - Objects: ACCESS_LOG, PERSONNEL, CERTIFICATION, LAB, CHEMICAL: Each access event has an access_type of 'Entry' or 'Exit' and an authorized_flag automatically determined by the trigger based on the Personnel member's valid Certification relative to the hazard_level of the related Lab/Chemical.
- BR15 - Objects: ACCESS_LOG, PERSONNEL, CERTIFICATION: The system does NOT allow an Access Log record to be inserted (INSERT) with authorized_flag = TRUE if the Personnel member does not have the corresponding valid Certification. This is the central DB Challenge of the project and will be implemented using an SQL Trigger in Phase 3.

2. Database Design

2.1 Conceptual Model

2.1.1 Comprehensive ER/EER Diagram



2.1.2 Entity Overview

The table below summarizes the main entities, their primary keys, and their purposes. Full details of data types and constraints will be presented in the Data Dictionary in Phase 2 (Logical Design).

Entity	Primary Key (PK)	Purpose
DEPARTMENT	dept_id	Department/unit responsible for managing faculty members.
PERSONNEL	person_id	Common supertype for all personnel (faculty, students, etc.).
FACULTY	person_id (FK)	Subtype: faculty member; may serve as a grant PI and advise students.
STUDENT	person_id (FK)	Subtype: student; may have a Faculty member as an advisor.
CERTIFICATION	cert_id	Safety certification type / authorized hazard-access level.
PERSON_CERTIFICATION	person_id, cert_id	N:N associative table recording certifications issued to each person and their validity.
LAB	lab_id	Laboratory with its own hazard level.
EQUIPMENT	equipment_id	Equipment assigned to a specific Lab.
CHEMICAL	chemical_id	Chemical stored in a Lab, with a hazard_class.

RESEARCH_GRANT	grant_id	Research grant with one PI (Faculty).
GRANT_MEMBER	grant_id, person_id	N:N associative table between a Grant and participating Personnel, including roles.
ACCESS_LOG	log_id	Lab entry/exit log with optional Equipment/Chemical references and an authorized_flag determined by a trigger.

2.1.3 Relationship Summary

Relationship	Cardinality (min,max)	Related Business Rule
DEPARTMENT - FACULTY	1,1 - 0,N	BR2
FACULTY - STUDENT (advises)	0,1 - 0,N	BR3
PERSONNEL - PERSON_CERTIFICATION - CERTIFICATION	1,1 -(0,N)-(1,N)- 1,1	BR4, BR5
LAB - EQUIPMENT	1,1 - 0,N	BR6
LAB - CHEMICAL	1,1 - 0,N	BR7
FACULTY - RESEARCH_GRANT (is PI of)	1,1 - 1,N	BR9, BR10
RESEARCH_GRANT - GRANT_MEMBER - PERSONNEL	1,1 -(1,N)-(0,N)- 0,1	BR11
PERSONNEL - ACCESS_LOG	1,1 - 0,N	BR13
LAB - ACCESS_LOG	1,1 - 0,N	BR13
EQUIPMENT - ACCESS_LOG (optional)	0,1 - 0,N	BR13
CHEMICAL - ACCESS_LOG (optional)	0,1 - 0,N	BR13
PERSONNEL -(d)- FACULTY / STUDENT	EER disjoint, partial specialization	BR1

2.1.4 Design Assumptions

- A Personnel member cannot be both Faculty and Student at the same time (disjoint specialization).
- A Grant has only one PI at a time; if the PI changes, pi_id is updated (the PI change history is not stored in Phase 1).
- Each Access Log record is associated with at most one Equipment item OR one Chemical (neither is mandatory), supporting detailed traceability in accordance with BR13.
- The hazard level (hazard_level/hazard_class) uses a consistent categorical scale across Lab, Chemical, and Certification (for example: None, Low, Medium, High) so that the trigger in Phase 3 can compare them directly.

2.2 Logical Schema Mapping

2.3 Normalization Verification